Part III Quantum Field Theory

Introduction

This PDF is a transcription of my handwritten notes for the Part III Quantum Field Theory course offered during Michaelmas term 2025 at the University of Cambridge. This course was lectured by Alejandra Castro but any typos or mistakes in these notes are strictly *my fault*! These notes are organised lecture–wise.

Lecture 1

A major theme of 20th century physics was the construction of physical theories which are local. Locality emerges naturally in so-called *field theories*, of which Maxwell's electrodynamics and Einstein's theory of general relativity are classical examples. One motivation for quantum field theory (QFT) is to formulate quantum mechanics in a way which is compatible with notions of locality, as described by Einstein's relativity.

In this course, we work with the natural unit convention; space and time have equal dimension (meaning that speeds are dimensionless). In particular, the speed of light in vacuum is set to one $c = 1$. Also, we set the reduced Planck constant equal to one $\hbar = 1$. Setting $\hbar = 1$ in addition to $c = 1$ implies that energies have dimension equivalent to an inverse length (inverse time). As is standard in particle physics, we use a mostly–minus signature for the Minkowski metric, $\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$.

review of classical field theory

It's well-known that classical mechanics can be formulated in terms of Hamilton's variational principle. If our system is described at time t by generalised coordinates $q = (q_1, q_2, \ldots)$ and generalised speeds $\dot{q} = (\dot{q}_1, \dot{q}_2, \ldots)$, we suppose the existence of a so-called *Lagrangian* function $L(q, \dot{q}; t)$ which encodes all dynamics. In particular, we define the *action* to be the temporal integral

$$S_{[q]} = \int_{t_i}^{t_f} dt L(q, \dot{q}; t),$$

where it is understood that the action S is a functional of the trajectory q. Hamilton's variational principle states that the physical trajectory $\tilde{q}$ extremises the action

$$\left.\frac{\delta S}{\delta q}\right|_{q=\tilde{q}} = 0.$$

The standard way to pose this problem is to say, given some Lagrangian $L(q, \dot{q}; t)$ and boundary conditions $q(t_i) = q_i$ and $q(t_f) = q_f$, how does one find the physical trajectory $\tilde{q}$? The standard way to answer this question is to write $q = \tilde{q} + \delta q$ where δq is, by some standard, a small deformation about the extremal trajectory which respects boundary conditions ($\delta q(t) = 0$ at $t = t_i, t_f$). By expanding the Lagrangian to first order in δq we see

$$\delta S = \int_{t_i}^{t_f} dt \left\{ \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right\}.$$

Note that we've restricted our attention to Lagrangians which depend at most on *first* derivatives of the generalised coordinates. It's perfectly valid to study Lagrangians depending on higher–order derivatives of q but these yield equations of motion which are strictly more than second–order in the coordinates – something physicists traditionally avoid. There is an implied summation over all the coordinate indices (i.e. you should read $\partial L/\partial q \times \delta q$ as $\sum_i \partial L/\partial q_i \times \delta q_i$ and so on). Noting that $\delta \dot{q} = d\delta q/dt$, we apply the product rule to find

$$\delta S = \int_{t_i}^{t_f} dt \left\{ \frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) \right\} \delta q + \frac{\partial L}{\partial \dot{q}} \delta q \bigg|_{t_i}^{t_f},$$

where the boundary term on the right vanishes by our assertion that $\delta q(t) = 0$ when $t = t_i, t_f$. Hamilton's principle is then satisfied iff the so-called Euler-Lagrange equations hold true

$$\frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0.$$

By substituting our particular Lagrangian into the EL equation and simplifying, we obtain the system's so-called *equations of motion*.

In the next lecture, we adapt this variational framework to a field degree–of–freedom (i.e. we treat the generalised coordinate as a function ϕ on Minkowski spacetime). This involves increasing the number of parameters integrated over in the action from one (just the time) to four (time + three spatial dimensions). Application of Hamilton's variational principle will then produce EL equations which constrain the spacetime evolution of the field.

Importantly, the symmetries of the system are reflected in the symmetries of the action. If some transformation of the system leaves the action unchanged up to some total derivative, the resulting EL equations (hence the equations of motion) remain invariant. This is what it means for a particular transformation to be a symmetry of our model. In quantum field theory, we assert that all Lagrangians must transform as scalars under Lorentz transformations (the group of rotations + boosts).